

A Bio-Composite Asphalt Binder for Advancing Sustainability in Pavement Engineering

Akram Hossain

Master of Science Thesis
Stockholm, Sweden
2025

TRITA-ABE-MBT-26503

Akram Hossain (akramh@kth.se)

KTH Royal Institute of Technology

School of Architecture and the Build Environment

Department of Civil and Architectural Engineering

Division of Highway and Railway Engineering

Stockholm, Sweden, 2025

Abstract

This study investigates the bio-composite asphalt binders performance in terms of mechanical and environmental impacts in pavement engineering towards sustainability. the composite asphalt binders that contain the reclaimed asphalt pavement (RAP) with 5% tall oil and demonstrating transparent comparison with base bitumen (70/100 grade) and RAP alone. The tall oil is a bio-based rejuvenator that comes from the Kraft pulping byproducts. The aged RAP binder becomes stiffer and brittle due to oxidative aging which limits its reuse potential without modification. Tall oils have fatty and rosin acid components that contribute to maltene replenishment and improved aggregate adhesion. Those actions give the composite asphalt binders lower stiffness and a better viscoelastic balance.

Rheological evaluation is performed using a Dynamic Shear Rheometer (DSR) with consider low to high temperature and fixed frequency sweeps as per AASHTO Standard. A master curve is constructed by time-temperature superposition as an analysis outcome. Moreover, Cole–Cole plot analyses demonstrated that tall oil-modified RAP binders achieve improved compliance at low frequencies and temperatures while maintaining adequate high-temperature rutting resistance.

Incorporation of 5% tall oil into asphalt binder modifies its rheological properties by disrupting polar asphaltene clusters and restoring aromatic and resin fractions. This alteration leads to an enhanced viscoelastic balance in the binder. These changes are evidenced by Dynamic Shear Rheometry (DSR) tests, master curve construction and Cole–Cole plot representations which together demonstrate improved deformation resistance and elastic recovery characteristics at service temperatures.

The integration of oil into RAP binders reduces the need for petroleum binder and uses an industrial byproduct. The reduction in petroleum binder offers environmental benefits. Furthermore, the use of oil contributes to economic advantages because the pavement lasts longer and uses resources more efficiently.

Considerations for industrial scalability, the technology needs temperature control, sufficient shear mixing, a clear supply chain and a strong quality assurance.

Nevertheless, future research needs are also highlighted, which include optimization of tall oil dosage, synergistic combinations with other bio-based additives and long-term field performance study to confirm laboratory observations and support specification development for sustainable pavement infrastructure.

Keywords: Bio-based rejuvenator, Rheology, Tall oil, Master curve, Cole-cole diagram, Sustainable asphalt.

Sammanfattning

Denna studie undersöker prestandan hos biokomposit-asfaltbindemedel ur både mekanisk och miljömässig synvinkel inom väg- och beläggningsteknik med fokus på hållbarhet. De komposita asfaltbindemedlen innehåller återvunnen asfalt (RAP) med 5 % tallolja och jämförs på ett tydligt sätt med basbitumen (penetrationsklass 70/100) samt RAP utan tillsats. Tallolja är en biobaserad föryngrare som kommer från biprodukter vid Kraft-massaprocessen. Det åldrade RAP-bindemedlet blir styvare och sprödare på grund av oxidativ åldring, vilket begränsar dess återanvändningspotential utan modifiering. Tallolja innehåller fettsyror och hartsyrakomponenter som bidrar till att återställa maltenfraktionen och förbättra vidhäftningen mot ballast. Dessa effekter ger de komposita asfaltbindemedlen lägre styvhet och en bättre viskoelastisk balans.

Den reologiska utvärderingen genomförs med en dynamisk skjuvreometer (DSR) med hänsyn till låga till höga temperaturer samt frekvenssvep vid fasta förhållanden enligt AASHTO-standard. En masterkurva konstrueras genom tids–temperatur-superposition som ett analysresultat. Vidare visade Cole–Cole-diagramanalyser att talloljemodifierade RAP-bindemedel uppnår förbättrad följsamhet vid låga frekvenser och temperaturer, samtidigt som tillräcklig spårbildningsresistens vid höga temperaturer bibehålls.

Inblandning av 5 % tallolja i asfaltbindemedlet modifierar dess reologiska egenskaper genom att störa polära asfaltenkluster och återställa aromatiska och resinfraktioner. Denna förändring leder till en förbättrad viskoelastisk balans i bindemedlet. Förändringarna påvisas genom DSR-tester, konstruktion av masterkurva och Cole–Cole-diagram, vilka tillsammans visar förbättrad deformationsresistens och elastisk återhämtning vid brukstemperaturer.

Integreringen av olja i RAP-bindemedel minskar behovet av petroleum-baserat bindemedel och utnyttjar en industriell biprodukt. Minskningen av petroleum-baserat bindemedel ger miljöfördelar. Dessutom bidrar användningen av olja till ekonomiska fördelar eftersom beläggningen får längre livslängd och resurser används mer effektivt.

Med avseende på industriell skalbarhet kräver tekniken temperaturkontroll, tillräcklig skjuvblandning, en tydlig försörjningskedja samt en stark kvalitetssäkring.

Trots detta lyfts även behov av framtida forskning fram, inklusive optimering av talloljedosering, synergistiska kombinationer med andra biobaserade tillsatser samt långsiktiga fältprestandastudier för att bekräfta laboratorieobservationer och stödja utvecklingen av specifikationer för hållbar beläggningsinfrastruktur.

Nyckelord: Biobaserad föryngrare, Reologi, Tallolja, Masterkurva, Cole–Cole-diagram, Hållbar asfalt.

Acknowledgements

Completing this master's thesis at KTH has been an incredibly rewarding journey filled with learning and collaboration on sustainable asphalt binders. I am deeply grateful to everyone who contributed to its success, from providing guidance and expertise to offering unwavering support throughout this process.

First and foremost, I would like to express my deepest gratitude to my teacher and thesis supervisor, Prof. Dr. Maria Chiara Cavalli for her continuous support and invaluable guidance throughout this work. Her comprehensive knowledge and expertise in bio-composite materials and sustainable pavement engineering played a key role in shaping this thesis. I am especially grateful for her help with preparing the samples and collecting those from the source. This work would not have been possible without the insight and experience she so generously shared with me.

I would also like to give special thanks to doctoral student Alexandra Kuksova for her outstanding support during the laboratory phase. Her assistance with laboratory testing at the KTH Laboratory and her guidance from the very beginning to the final steps of the experimental work. Her engagement was essential to completing this thesis. Her advice, suggestions, and encouragement were incredibly helpful and contributed greatly on this thesis.

In addition, I also thank my classmates and friends for the pleasure of their collaboration. Their practical help and emotional support provided a much-needed boost during this process.

Finally and most importantly, I would like to express my heartfelt gratitude to my parents. Even though they are far away in my home country, their constant encouragement, trust and support have always been my greatest source of motivation. Their strength has been with me at every step of this academic journey.

Contents

Chapter 1 12

1. 12

1.1 12

1.2 13

1.3 13

1.4 13

1.5 14

Chapter 2 16

2 16

2.1 16

2.2 16

2.2.1 16

2.2.2 17

2.2.3 19

2.3 20

2.3.1 20

2.3.2 21

2.3.3 22

2.4 23

2.4.1 23

2.4.2 23

2.4.3 24

2.5 26

2.6 26

Chapter 3 28

3 28

3.1 28

3.2 29

3.3 30

3.3.1 30

3.3.2 31

3.4 32

3.5 33

3.6 33

Chapter 4 34

4 34

4.1 34

4.2 35

4.3 38

4.4 40

Chapter 5 42

5 42

5.1 42

5.2 43

Bibliography 45

List of Figures

Figure 1: DSR Testing Set-up (Source: Yusoff, N.I.M. et al., 2013) 21

Figure 2: DSR testing specimens 29

Figure 3: DSR Testing Preparation in KTH Laboratory Instrument 30

Figure 4: Master Curve for individual binders 37

Figure 5: Cole-Cole Diagram of different binders (Log Scale) 39

Figure 6: Cole-Cole Diagram of different binders (Non-Log Scale) 40

List of Tables

Table 1: Key Industrial Advancements and Citations 18

Table 2: Key Drivers in Recycling Evolution 20

Table 3: Comparative Material Properties 28

Table 4: Considered Frequency Interval for Testing 31

Table 5: Parameter values used for all binders in WLF formula and Sigmoidal model 34

Table 6: Observed stiffness of all binders 36

Table 7: Binder Cole-Cole Diagram Curve Characteristics 39

Abbreviations

BB	Base bitumen
TO	Tall oil
RAP	Reclaimed asphalt pavement
DSR	Dynamic shear rheometer
SARA	Saturates, aromatics, resins, asphaltenes
WLF	Williams-Landel-Ferry
TTSP	Time-Temperature Superposition Principle
LVE	Linear viscoelastic

Chapter 1

1. Introduction

1.1 Background

The global emphasis on sustainable development has forced the search for greener alternatives to traditional materials in civil engineering, particularly pavement construction. Traditional asphalt binders made from petroleum have substantial environmental consequences, including greenhouse gas (GHG) emissions and dependency on nonrenewable resources (Chen et al., 2023). The construction industry is under increasing pressure to match global sustainability targets such as those set in the United Nations Sustainable Development Goals (SDGs). Consequently, there is an urgent need to explore alternative binders that align with environmental objectives while maintaining or improving pavement performance. To address these challenges, researchers are exploring bio-based alternatives that integrate renewable and waste materials (Cavalli M.C. et al. 2024).

Bio-composite asphalt binders present a promising solution. These binders incorporate renewable materials like tall oil, a byproduct of the kraft process for wood pulp manufacture, primarily when processing coniferous trees such as pine (Aro T. et al. 2017). The use of bio-based components not only reduces reliance on non-renewable resources but also promotes circularity by repurposing industrial waste. Recent advancements in the treatment of reclaimed asphalt pavement (RAP) using bio-based rejuvenators further enhance their appeal, as they enable recycling of aged materials without compromising performance (Cavalli M.C. et al. 2024).

Despite advances in bio-based binders, there are issues such as inconsistency in raw material quality, ageing resistance and scalability limit of their wide adoption (Yu et al., 2014; Gao et al., 2025). To assure their suitability for large-scale applications, bio-composite asphalt binders must be thoroughly evaluated in terms of mechanical, rheological and environmental performance.

To validate bio-composite potential, rheological testing plays a critical role in evaluating their mechanical, thermal and ageing properties. For this case, studies have demonstrated that bio-rejuvenators can restore the rheological properties of aged binders, mitigating the challenges of oxidative ageing and cracking (Moins et al., 2022). This ensures that the binders meet the stringent demands of modern road systems. Additionally, life cycle assessments (LCA) are crucial to quantify their environmental benefits, including reductions in carbon emissions and energy consumption. By integrating laboratory testing with environmental analyses, this project aims to advance the adoption of bio-composite asphalt binders as a cornerstone of sustainable infrastructure (Gao J. et al., 2025; Cavalli M.C. et al. 2018).

1.2 Purpose of Study

The purpose of this study is to determine the viability of bio-composite asphalt binders as a sustainable alternative to conventional petroleum-based binders in pavement engineering. This study addresses rheological performances of bio-based reclaimed asphalt pavement binders. The study's goal is to comprehensively evaluate use in reclaimed asphalt pavement.

1.3 Study Aim

The fundamental aim of this study is to determine whether bio-composite asphalt binders that have been modified using tall oil and RAP binders can effectively replace traditional binders. The study specifically aims to:

- Evaluate the mechanical and rheological properties of bio-composite asphalt binders compared with conventional asphalt binders.
- Evaluate the effectiveness of bio-composites in rejuvenating ageing asphalt materials like RAP.

1.4 Methodology

In order to achieve the aim of this study, first a literature review is performed. After studying existing research and analysis, this study has employed a methodology to evaluate bio-composite asphalt binder feasibility, performance and sustainability. The approach includes material preparation and rheological testing in laboratory with specific equipment.

- **Material Preparation:** 5% tall-oil is blended with base bitumen to create bio-bitumen binders. Laboratory tests are conducted to ensure the homogeneity and stability of the blend.
- **Rheological Testing:** Dynamic Shear Rheometer (DSR) is used to assess the viscoelastic properties under varying temperatures and loading conditions. Representing the Master curve and Cole-Cole graph for all three binders (Rap 5% tall oil, Base Bitumen 70/100 & RAP), whereas the master curve represents the complex modulus (G*) as a function of frequency at a reference temperature obtained via time-temperature superposition. A Cole-Cole graph showing the relationship between the elastic (storage modulus, G′) and viscous (loss modulus, G″) components illustrates how a material behaves both elastically and viscously. (Gao J. et al., 2025; Cavalli M.C. et al. 2018; Nazzal, M.D., et al., 2014).

1.5 Outline

The outline of all chapters provides an overview of this master's thesis structure as follows:

Chapter 1: Introduction – Provides the background, purpose, aims and methodology of the study.

Chapter 2: Literature Review – presents reviews of existing research on bio-composite binders, focusing on rheological properties, RAP rejuvenation and environmental impacts, drawing from key studies such as Cavalli M.C. et al. (2018 & 2024); Chen et al. (2023) and Gao et al. (2025). By providing a comprehensive overview of existing research on bio-based asphalt binders, this study has been critically analysed to identify knowledge gaps and establish the theoretical foundation for this work. At the beginning of the chapter, information about Reclaimed Asphalt pavement (RAP) and reasons for rejuvenating this is discussed, taking into consideration both technical and economic aspects. As tall oil has been applied as a rejuvenator, a short description of its benefits and source is discussed. Then introduce the samples and their testing methods in the laboratory and illustrate the provable outcome based on supporting literature. Further, based on technical and economic analysis conducted in the past, the environmental and economic effect of using RAP with virgin asphalt in highway construction is considered.

Chapter 3: Experimental Methodology – This chapter outlines is details of the experimental design for developing and testing the bio-composite asphalt binder. It includes material preparation and a comprehensive overview of rheological testing using DSR. After completion

of the laboratory testing, a concise environmental and economic benefits is demonstrated based on the peer reviewed literature since rheological aspects for sustainability is forced on this study.

Chapter 4: Results and Discussion – This chapter presents the findings from DSR testing, the rheological behavior of the bio-composite binder. Comparative analysis with conventional binders has been conducted between the results and discussed the possible outcomes. The results in the context of sustainability goals and practical implementation of the binders. It also addresses challenges such as cost-effectiveness and environmental benefits.

Chapter 5: Conclusion and Recommendations – Summarises key results of the study which discusses implications of RAP binders and minimizes the conventional binder's uses for pavement engineering. Moreover, this section summarises the contributions of this research to pavement engineering and sustainability. The author offers recommendations for future studies to improve existing knowledge about bio-based asphalt binders and their real-world applications along with advantages based on the thesis.

Chapter 2

2 Literature Review

2.1 Context

The asphalt paving industry faces increasing pressure to reduce its environmental footprint whilst maintaining or improving pavement performance. Two parallel trends have emerged: the increased utilization of Reclaimed Asphalt Pavement (RAP) to promote circular economy principles and the incorporation of bio-based modifiers derived from renewable feedstocks to partially replace petroleum-based binders (Fini et al., 2012). The convergence of these approaches by combining RAP with bio-additives represents a promising pathway towards sustainable pavement engineering.

Bio-oils and bio-binders can be derived from various renewable sources including agricultural residues, forestry byproducts, and industrial waste streams. Tall oil, a byproduct of the Kraft pulping process, represents one such industrial waste stream with potential application in asphalt modification. However, the specific performance characteristics of tall oil in combination with RAP require careful evaluation through standardized laboratory testing protocols.

2.2 Historical Development of Asphalt Binders

2.2.1 Natural Bitumen in Ancient Constructions

The first historical use of natural bitumen for building purposes dates back to at least 7th century BC in Babylon where it was extracted from naturally occurring sources. It was used for its adhesive and hydraulic properties in masonry and hydraulic facilities (Cavalli, 2018). When placed between stone or baked clay bricks, bitumen provided joints that are lasting and not waterproof which prevented water from wetting-and-drying the buildings over time in a saturated environment (Cavalli, 2018). In ancient hydraulic systems, it was also suitable for roads, irrigation channels, lining reservoirs and canals to prevent water seepage due to its hydrophobic and viscoelastic properties as well (Cavalli, 2018). Parallel uses have also been identified at other Mesopotamian sites, in the Levant, in Persia, in Egypt and the Indus valley

(e.g., for ship caulking as well as water-preserving public construction), indicating its early use in infrastructure and hygiene (Cavalli, 2018). The quality of the ancient bitumen also depended on the geological origin; deposits with a higher content of maltenes yielded softer and more elastic binders. While sources richer in asphaltenes presented stiffer materials with less elasticity that could easily crack, findings which can be correlated to contemporary issues regarding rheology and performance grading of the binder (Cavalli, 2018). Modern petrographic results on the ancient bitumen samples can characterize long-term oxidation and microstructural changes. These are valuable analogs for understanding the aging mechanism of modern RAP binders as well as further confirming the continuity between past and present research on asphalt durability (Cavalli, 2018).

2.2.2 Industrial Advancements in Asphalt Production

The evolution from natural bit to modern engineered asphalt binders has been driven by major industrial advances in processing and mix design, aimed at meeting growing traffic demands and ensuring durability under diverse climates and loading conditions. Early petroleum refining used distillation to obtain more uniform binders from crude oil, allowing control over viscosity and softening point so that asphalt could be tailored to regional climate needs (Sani, Sutanto, and Setiadji, 2025). Subsequent fractionation techniques enabled systematic separation of maltenes and asphaltenes, improving the predictability of mechanical performance. A key step forward is treating Reclaimed Asphalt Pavement (RAP) as an engineered constituent rather than waste which led to plant modifications to prevent excessive aging during production and to support higher RAP contents without sacrificing workability or cracking resistance (Sani, Sutanto, and Setiadji, 2025; Elnaml et al., 2024). In parallel, binder chemistry advanced with the development of polymer-modified asphalt (PMA), incorporating polymers such as styrene-butadiene-styrene (SBS) and ethylene-vinyl-acetate (EVA) to enhance viscoelastic recovery and performance under heavy loads and extreme temperatures (Gao et al., 2025). Computational tools such as Density Functional Theory (DFT) began to support formulation work by predicting compatibility issues and guiding additive design to reduce oxidative and other forms of degradation (Fini and Hajikarimi, 2025c). Sustainability has become increasingly central with byproducts like tall oil soap redirected from other industrial waste streams into asphalt binder modification that reduces reliance on petroleum-based inputs (Cavalli, Wu and Poulikakos, 2024). Warm Mix Asphalt (WMA) technologies further

contributed by lowering production temperatures through foaming and chemical additives. It is cutting fuel and emissions while maintaining performance and particularly benefiting mixtures with bio-based modifiers (Sani, Sutanto, and Setiadji, 2025). At the same time, modern quality control has adopted advanced rheological tests such as dynamic shear rheometer measurements to ensure mixtures meet performance criteria and to optimize the balance between high RAP usage and resistance to rutting and other distresses (Elnaml et al., 2024; Toth et al., 2023). Collectively, contemporary asphalt production reflects a convergence of chemical innovation, process engineering and data-informed design. These are enabling more sustainable, high-performance pavements that align with increasing infrastructure and environmental requirements (Cavalli, Wu and Poulikakos, 2024; Fini and Hajikarimi, 2025c). In Table 1, a summary of key industrial advancements is demonstrated with citations.

Table 1: Key Industrial Advancements and Citations

Advancement Area	Description	Key Citations
Raw Material Processing	Transition to distillation/fractionation for uniform binders.	Sani, Sutanto, and Setiadji (2025)
Reclaimed Materials	Integration of RAP requires plant modifications to prevent aging.	Sani, Sutanto, and Setiadji (2025); Elnaml et al. (2024)
Binder Modification	Polymer-modified asphalt (PMA) using SBS/EVA for durability.	Gao et al. (2025)
Computational Tools	Use of Density Functional Theory (DFT) for formulation prediction.	Fini and Hajikarimi (2025c)
Sustainability	Recycling tall oil soap and implementing Warm Mix Asphalt (WMA).	Cavalli, Wu, and Poulikakos (2024); Sani, Sutanto, and Setiadji (2025)
Quality Control	Advanced rheological testing to ensure rutting resistance.	Elnaml et al. (2024); Toth et al. (2023)

2.2.3 Evolution of Asphalt Recycling Practices

The evolution of asphalt recycling has transitioned from the opportunistic reuse of milled materials to a strategically engineered process shaped by environmental policies and technological advancements. Initially, low fractions of reclaimed asphalt pavement (RAP) have been incorporated chiefly for cost savings, leading to underutilization of the material since older binders could result in stiffer mixtures (Tarsi et al., 2020). As the costs of virgin materials increased, innovations such as parallel drum systems have been developed. These are allowing for higher RAP content while minimizing thermal exposure to prevent the degradation of binder quality (Elnaml et al., 2024). Subsequent research highlighted the significance of blending aged RAP binders with new virgin binders. It's revealing that excessive RAP without proper rejuvenation heightened stiffness which could adversely affect fatigue life (Tarsi et al., 2020).

Consequently, the introduction of bio-based rejuvenators provided an avenue to improve flow ability and fatigue resistance while maintaining environmental standards (Zhang et al., 2020). Regulatory frameworks initially set limits on RAP inclusion rates due to quality concerns. However, this inadvertently increased stockpiles and spurred rigorous testing to ensure mechanical compatibility in high-RAP blends (Tarsi et al., 2020). Warm Mix Recycled Asphalt (WMRA) technologies have emerged and allowing producers to lower mixing temperatures and subsequently preserve rejuvenator quality during asphalt production (Wang et al., 2020). In regions like South Africa, guidelines permit significant RAP use while requiring specific characterizations of blended materials, reinforcing the need for systematic mix design adaptations (Tarsi et al., 2020).

Economic factors also play a crucial role; while reclaimed asphalt reduces material costs, bio-based rejuvenators often entail higher initial expenses but can offer long-term savings by enhancing performance and durability (Cavalli, Wu and Poulikakos, 2024). The ongoing integration of lifecycle assessments is coupled with improved stockpile management and underscores a shift towards optimizing technical recycling processes. As a vital engineered component in modern asphalt mixtures, it is also facilitating a move away from treating reclaimed asphalt as mere filler toward recognizing it (Elnaml et al., 2024). Table 2 summery up the recycling evolution with citations.

Table 2: Key Drivers in Recycling Evolution

Factor	Impact on Practice	Citation
Economic Pressure	Rising virgin material costs drove the development of systems for higher RAP content.	(Elnaml et al., 2024)
Technical Innovation	Bio-based rejuvenators and WMRA improved fatigue resistance and lowered production temps.	(Zhang et al., 2020); (Wang et al., 2020)
Regulatory Changes	Shift from strict caps to characterization-based guidelines to manage stockpiles.	(Tarsi et al., 2020)
Lifecycle Analysis	Focus shifted to long-term durability and environmental assessment rather than just upfront cost.	(Cavalli, Wu, and Poulikakos, 2024)

2.3 Rheological Evaluation Methods

2.3.1 Dynamic Shear Rheometer (DSR) Principles

The Dynamic Shear Rheometer (DSR) is one of the most common devices for the performance grade of asphalt binders since it measures the viscoelastic behavior of binder materials at controlled temperature and loading conditions. The function is to apply oscillatory shear stress to simulate pavement traffic stresses, testing resistance to deformation as per AASHTO T-315 standard. Due in part to the elastic and viscous behavior of asphalt binders, DSR has been used to show the susceptibility of an asphalt mix when subjected to a sinusoidal shear load on a thin sample between parallel plates by quantifying it as the resistance to deformation under oscillatory load that being $G*$ complex modulus (Cavalli et al., 2018). This modulus can be divided into the storage modulus G', representing the elastic (energy storing) portion and loss modulus G'', reflecting viscous (dissipation energy) portion of the binder material (Cavalli et al., 2018). The phase angle δ reflects the equilibrium between these contributions and determines whether the binder is more elastic or more viscous ($0° < \delta < 90°$), which depending on its temperature and age (Fuhaid et al., 2018). Rheological behavior measurements are very

sensitive to small temperature variations and testing protocols as per AASHTO T315 confirm the importance of strict temperature control. At high service temperatures, DSR is mainly used for evaluating rutting susceptibility since stiffness and load-carrying ability are the major concerns. Moreover, assessing the effect of additives based on their impact on modulus-frequency relationship (Guo et al., 2018). At lower or moderate temperatures, fatigue-related protocols such as the Linear Amplitude Sweep (LAS) testing protocol are used to characterize cracking resistance which is particularly crucial for mixtures containing aged RAP binders (Xu et al., 2023). Pavement distress mechanisms such as rutting and fatigue cracking can be characterized using performance indices calculated from DSR parameters $G*/\sin\delta$ (higher value suggests better rutting resistance) and $G*\cdot\sin\delta$ (lower values indicate better fatigue resistance), thereby providing a direct correlation between rheological properties of the binder and pavement performance (Fuhaid et al., 2018). As measurement accuracy depends on the equipment setup, careful calibration and sample preparation are necessary, especially for testing stiff RAP binders, which require careful handling and conditioning (Barco Carrión et al., 2023). In general, because microscale viscoelastic properties dictate field performance, DSR is an essential tool in contemporary asphalt binder characterization and mixture design (Cavalli et al., 2018; Xu et al., 2023). Figure 1 provides a simple illustration of DSR testing method.

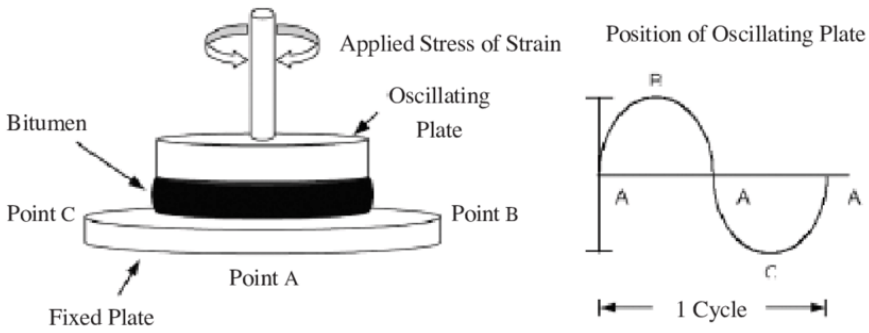


Figure 1: DSR Testing Set-up (Source: Yusoff, N.I.M. et al., 2013)

2.3.2 Master Curve Analysis in Binder Studies

Master curve analysis is a critical tool in binder rheology as it allows viscoelastic behavior over broad frequency ranges to be summarized using the Time-Temperature Superposition Principle (TTSP) (Imaninasab et al., 2024). This method is particularly advantageous for complex materials such as RAP containing binders rejuvenated with bio-based additives which often are temperature sensitive and have heterogeneous aging levels. Thereby, the geographically or

seasonally distributed sets of modulus-frequency data are all transformed to a common reference temperature. Thus, the time-temperature master curve condenses the geographic or seasonal variation into one consolidation framework which can be used for predicting its performance. The two key factors include dynamic modulus G∗ and phase angle δ obtained from the frequency sweep tests (Pei and Fan, 2021). This detailed testing is essential to assess whether modifications to the binder maintain a desired ratio of elastic to viscous behaviour over a range of service conditions. Although master curves for bio-oil RAP binders demonstrate lowered complex modulus, typical of increased flexibility, some petroleum softeners could uniformly reduce stiffness and sacrifice high-temperature behavior. The 2S2P1D model is a popular tool for mathematically describing master curve data by stipulating binder responses in terms of different viscous and elastic elements. Prominent departures from thermo rheological simplicity in TTSP shifting, particularly for some specific warm-mix additives, signal distinct temperature effects on relaxation mechanisms that can impact performance (Xu et al., 2023). Furthermore, inclusion of information provided by phase angle, master curves in the binder behaviour yields a more thorough understanding. It will allow us to anticipate performance limits for specific additive chemistries under real loading environments (Imaninasab, R., Loria-Salazar, L. and Carter, A., 2024; Guo et al., 2024).

2.3.3 Cole-Cole Plots in Viscoelastic Characterization

The Cole–Cole plots are also valuable methods for illustrating viscoelastic nature of the asphalt binders by plotting loss modulus G″ against storage modulus G’ without any time–temperature superposition shifting (Toth, Petho and Rosta, 2023). This method offers the possibility to directly assess this type of shifting behavior from elastic to viscous domain, where the left side of the plot stands for dominance in viscosity whereas right part displays performance in elasticity (Guo et al., 2018). The profile of the curves is a qualitative indication of compatibility as well. With symmetric ones indicating good dispersion in more compatible systems, for instance, those containing hydrophobic nanosilica and asymmetric ones suggesting poor molecular integration (Guo et al., 2018). Such plots are often combined with Black diagrams to expose weak additive effects on the energy distribution that may be difficult to detect otherwise (Wang et al., 2019). Besides, they can help analyze phase change materials and crystalline wax inclusions by emphasizing their impact on G′/G″ profiles (Wu and Hou, 2024).

Furthermore, this evolution of the curves past a period representing their aging conditioning, for example RTFO or PAV. Those are also very important to detect performance problems and trends related to the binder in aging (Barco Carrión et al., 2023). Overall, the Cole–Cole plots are powerful qualitative and semi-quantitative techniques for becoming better at additive criteria and mix quality in the processing of pavement materials design.

2.4 Materials Description

2.4.1 Base Bitumen Properties (70/100 Grade)

In the present work, the reference binder is virgin 70/100 penetration-graded bitumen. The penetration of the binder under European Standard (EN) 1427 is approximately 67×10^{-1} mm at 25°C and a Ring & Ball softening point of around 57°C. These properties situate the binder within a stiffness range suitable for climates with moderate seasonal variation. Also, it is providing a necessary rheological balance of elastic recovery and viscous flow to accommodate traffic loads without succumbing to excessive rutting or brittleness (Cavalli M.C. et al. 2018).

From a structural standpoint, the colloidal arrangement wherein maltenes act as the continuous phase and asphaltenes form the dispersed network significantly governs the binder's modulus-phase behavior under dynamic shear rheometer testing. Typical phase angles for the 70/100 grade fall between 40∘ and 55∘ across standard service temperatures. This temperature range is allowing for beneficial viscous contributions while preserving the elasticity required for load redistribution during stress cycles (Gowda et al., 2024). Chemically, the abundance of long-chain hydrocarbons and minimal polar functionality supports compatibility with both petroleum-derived and bio-based additives which is vital when integrating agents like tall oil that contain polar rosin acids. Furthermore, in rheological assessments using frequency sweeps, the unaged 70/100 base bitumen acts as a control curve. Its G∗ values generally remain elevated against aged RAP composites, underscoring its structural integrity and rutting resistance essential for applications in hotter climates.

2.4.2 RAP Binder Characteristics

The properties of recycled RAP binders mainly depend on environmental and earlier service conditions. Oxidative ageing induces important compositional variations, the main one being volatile depletion and asphaltene increase by polymerization (Cavalli, 2018). This chemical shift results in the increase of more polar species which results in the higher stiffness and typical

brittleness (Yaseen et al., 2022). Rheological measurement shows that although an increase in the complex modulus of G∗ improves rutting resistance, a decline in the phase angle δ weakens the viscous effect and renders viscoelastic materials susceptible to cracking (Gowda et al., 2024).

Additional testing also shows lower fracture toughness at low temperatures, suggesting that oxidative hardening is the main contributor to crack initiation (Cavalli, 2018). The steep stiffness slope of master curve analyses also indicates poor resistance to sustained loading, especially in the case of low-speed traffic (Xu et al., 2023). Structural non-uniformity is frequently encountered in Cole–Cole plots of the tensile specimens which indicates stiff clusters within the material (Guo et al., 2018). Furthermore, atomic force microscopy indicates the lower adhesive forces between RAP binding agents and aggregates compared with virgin materials which results in more severe moisture susceptibility (Nazzal et al., 2014). Ultimately, unrejuvenated RAP binders exhibit a natural response that is characterized by stiffness and viscoelastic disproportionality which typically requires rejuvenation treatments.

2.4.3 Bio-Composite Asphalt Binder Development

2.4.3.1 Formulation of RAP and Tall Oil Binder

The objective of its bio-composite asphalt binder that involves the use of tall oil is to improve the quality and sustainability aspects of recycled pavement materials. This process starts by obtaining tall oil, rich in fatty and rosin acids from the Kraft pulping byproduct stream and refining it through acidulation and purification before blending. A tall oil dosage of approximately 5% is applied on this study to rejuvenate aged RAP binders. Here fatty acids act as softening agents and rosin acids improve aggregate coating due to their polar characteristics (Aro and Fatehi, 2016). It is of paramount importance to maintain the correct storage and blending procedure to preserve tall oil functional properties, as a proper mixing at specific temperatures guarantees additive uniform dispersion (Cavalli et al., 2021).

The blending process includes the requirement of an accurate temperature and shear control to avoid volatilization losses and/or phase separation but also multiple stress creep recovery (MSCR) test. The test confirms mechanical and rheological targets after integration (Gowda et al., 2024). Microstructural compatibility is also confirmed by the Cole-Cole plot analysis, ensuring phase homogeneity (Guo et al., 2018). As a matter of industrial practice, tall oil can

be added to the roof coating materials by conventional asphalt plant-type mixers with only slight modification in plant operation.

From a chemical standpoint, tall oil rosin acids react with oxidized binder species to promote cohesion in the resulting asphalt which has economic and environmental pros: using waste industry products (it also comes from mechanical pulping) and we are emancipating ourselves from petroleum-based additives. For RAP + 5% tall oil binders, RAP binder composition and blending conditions should be taken into account as these have an impact on final mixture viscosity, flexibility and resistance to rutting (Zhu et al., 2025). Laboratory testing using Dynamic She Rheometer (DSR) provides early indicators of modified binder performance with attention given to harmonizing rheological properties across operational temperature ranges (Xu et al., 2023).

Overall, this approach combines scientific understanding of material interactions and meticulous process control to produce consistent, reliable and more sustainable asphalt binders suitable for diverse service environments. The performance of these binders will have mechanical integrity while taking full advantage of environmental benefits.

2.4.3.2 Interaction Mechanisms between Tall Oil and RAP Binder

The process of tall oil reacting with the aged binder in RAP is complex and results from the amphiphilic nature of tall oil molecules as well as colloidal imbalances present in oxidized asphalt. The "re-aged" of the RAP binders under oxidation leads to the development of higher asphaltene content and reduced aromatics which results in a stiff and brittle network (Cavalli, 2018). Tall oil acts with these long-chain fatty acids and resin acids to interfere with such dense formation, thereby locally recovering portions of mobility of the binder matrix. Fatty acids can migrate to the maltene phase and their polar head group contacts with asphaltenes, resulting in a decrease of interfacial tension and an increase of compatibility (Wu and Hou, 2024).

This chemical interaction weakens the intermolecular forces responsible for stiffness. This is also leading to a reduction in the complex modulus $G*$ during low-frequency testing without causing significant viscosity loss at higher frequencies (Xu et al., 2023). Resin acids can also provide steric stabilization on the surface of asphaltene particles which can alter the colloidal structure and enhance the flexibility of binder as a whole (Zhang et al., 2020).

Besides the rheological enhancements, this synergism enhances aggregate moisture proofing at its surfaces as indicated by increased tensile strength and cracking resistance upon freeze–thaw cycles (Wu and Hou, 2024). Nonetheless, to balance multiplicities of phase, concomitant careful control on thermodynamics is necessary. In this regard warm-mix procedures are required to keep volatile fatty acids conjointly essential for the miscibility. In the end, tall oil does more than just act as a diluting agent for stiffness. It also accomplishes an internal colloidal structure remodeling in such a way as to recover adhesion and performance, improvement that is truly optimized only by verifying binder shortages like SARA fractionation (Zhang et al., 2020).

2.5 Environmental Aspects Bio-based Binder

Evaluating the integration of 5% tall oil into RAP-rich asphalt mixtures requires a comprehensive life cycle perspective that accounts for environmental footprints and economic feasibility (Eurobitume, 2025). Frameworks like TRACI 2.1 allow the assessment of these bio-composites against traditional binder systems by measuring impacts like global warming potential (Elnaml et al., 2024). Although the use of RAP inherently saves materials, the inclusion of tall oil enhances such savings by replacing the petroleum-based rejuvenators with a renewable industrial coproduct (Tarsi et al., 2020). As tall oil is produced from waste streams, it provides an embedded environmental benefit with only small extra carbon emissions (Cavalli et al., 2024). In terms of production, using WMA technology with tall oil increases binder durability and reduces the global warming potential (Fini and Hajikarimi, 2025a).

2.6 Economic Aspects Bio-based Binder

Cost-benefit justification for the environmental and performance benefits of RAP with tall oil binder whether they will lead to sustainable economic advantage in the long term. In general, indicates that tall oil based bio rejuvenators are cost-competitive or even cheaper than conventional petroleum alternatives for high-RAP mixtures. Particularly if it is located near asphalt plants that minimize transport costs and emissions (Lv et al., 2021). While the refined tall oil may be more expensive to obtain as a result of additional processing, higher levels of uniformity in response to repeated or prolonged loading will reduce future maintenance (Fini and Hajikarimi, 2025b). During manufacturing, substituting tall oil in Warm-Mix Asphalt for the regular HMA (Hot Mix Asphalt) can lower mixing temperature which reduces energy and

oxidative aging of the binder, both enhancing durability and cost (Fini and Hajikarimi, 2025d). Comparison of the performance between tall-oil-treated RAP binders and untreated high-RAP mixes over pavement life is effective due to longer immune-protection time against tracking, fatigue, aging and some other distresses. Moreover, resurfacing prevention can achieve sustainable reduction in replacement cost for material, labor charge and traffic interruption (Wu and Hou 2024).

In particular studies on Life Cycle Cost Analysis indicates that the cumulative benefit to deferred maintenance over the life of a building can be big dollars. The recuperating action of tall oil also enhances ductility and allows higher RAP content in the future mixes. Therefore, improving recyclability as well as decreasing demand for virgin aggregates and binders (Tarsi et al., 2020). At a larger scale, it has been estimated that high adoption of RAP leads to potential savings in the billions of dollars each year as it decreases the use of freshly extracted materials (Tarsi et al., 2020). Especially regarding bio-based tall oil, the challenge lies on the uniform quality which may vary as per seasonal based on its composition and performance. Procurement quality requirements can mitigate this risk for a small cost (Fini and Hajikarimi, 2025d).

Under these technical and supply constraints, benefit–cost analysis in general projects attractive (i.e., positive) returns. Due to the energy savings largely and long life of RAP with 5% tall oil pavement which leads to better matching environmental benefits with strong economic benefits (Sani et al., 2025).

Chapter 3

3 Experimental Methodology

3.1 Materials properties

In this study, Base bitumen (70/100), RAP and RAP+5% Tall oil are used to evaluate and find the comprehensive advantages of bio-based rejuvenator. Base bitumen 70/100 is a conventional paving grade binder that offers a balanced viscoelastic behavior, providing sufficient resistance to rutting while maintaining enough flexibility to limit thermal and fatigue cracking in moderate climates. RAP is composed of milled or crushed fragments from existing asphalt pavements, containing aged binder and mineral aggregates. Due to long-term oxidative aging, the binder in RAP is significantly stiffer and more brittle than virgin bitumen. When 5% tall oil is added to RAP, it partially restores the aged binder's original rheological characteristics by increasing flexibility and reducing viscosity. The physical and chemical properties of these materials is summarizing in Table 3. Those listed properties are common but crucial for evaluating material performance and compatibility for the intended application.

Table 3: Comparative Material Properties

Property	Base Bitumen (70/100)	RAP	RAP + 5% Tall Oil
Appearance	Black, viscous liquid	Dark brown, thick, solid/paste-like mass	Dark brown, slightly less viscous than RAP
Particle size	Not applicable	Granular/milled asphalt fragments	Granular RAP coated with rejuvenated binder
Density (g/cm³)	1.01 – 1.06	1.0 – 1.2	1.0 – 1.2
Viscosity (Pa·s)*	Strongly temperature-dependent; moderate	High (stiff aged binder)	Lower than RAP due to rejuvenation
Water content (%)	Negligible	Typically ≤ 0.5	Negligible
Decomposition temperature (°C)	> 350	> 300	> 300
Auto-ignition temperature (°C)	~ 350	~ 300	~300
Softening point (Ring & Ball, °C)	≈ 57	≈ 65.7	Close to RAP, with improved flexibility
Flash point (°C)	≥225–230	250 – 300	250 – 300

3.2 Sample Preparation

Preparation of asphalt binder samples includes a systematic heating and during blending process to obtain a homogeneous mix between RAP-binder and tall oil additives. Virgin bitumen is heated to approximately 135°C to maintain workability while avoiding thermal degradation. For the RAP, the recovered RAP binder is first extracted using solvent methods and subsequently stripped of residual solvent through mild heating (Cavalli, 2018).

RAP and tall oil formation as described in Section 2.4.3. Tall oil should be stored and supplied at ambient temperatures before blending. Tall oil is incorporated through high-speed shear mixing at rates near 3000 rpm for over 30 minutes to ensure effective distribution of fatty acid and rosin components. The method of mixing RAP binders with tall oil is involved gradually adding oil which prevents uneven softening that could occur with direct addition methods (Wu and Hou, 2024). It is important to control the temperature during blending where working ranges go up to 165°C in order to ensure that complete dispersion are achieved without causing any mechanical breakdown of reactive components (Guo et al., 2024). This unified preparation approach allowed an improvement pathway towards using renewable binders in high-RAP

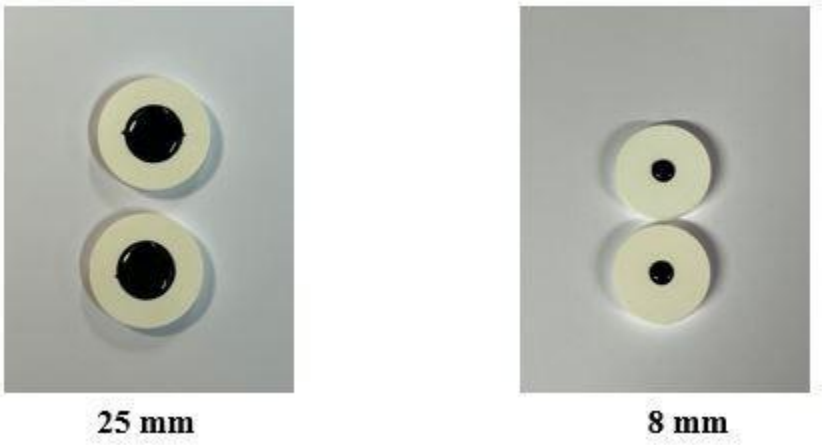


Figure 2: DSR testing specimens

asphalt mixtures on the industrial level to meet both research benchmarks and sustainable pavement development goals.

The specimens for rheological testing maintained consistent dimensions throughout the process. For high-temperature rutting tests, disc specimens are 25 mm in diameter with 1 mm thickness. And smaller 8 mm specimens with thicker diameter (2mm) are used for intermediate and low-

temperature testing sweeps. This standardized approach ensures reliable and comparable results across different temperature ranges during the rheological analysis (Gowda et al, 2024). Considering two samples for each diameter (Figure 2) and have taken average data for the resulting graph.

3.3 Dynamic Shear Rheometer Testing

Dynamic Shear Rheometer (DSR) testing includes both temperature sweep and frequency sweep methods. Temperature sweep testing measures changes in the binder's properties as temperature varies. Frequency sweep testing examines how the binder behaves under different loading frequencies (simulating various traffic conditions) while maintaining constant temperature. Both tests are important for fully understanding the material's performance. The test protocols for both test are performed as per AASHTO 315-08 guideline.

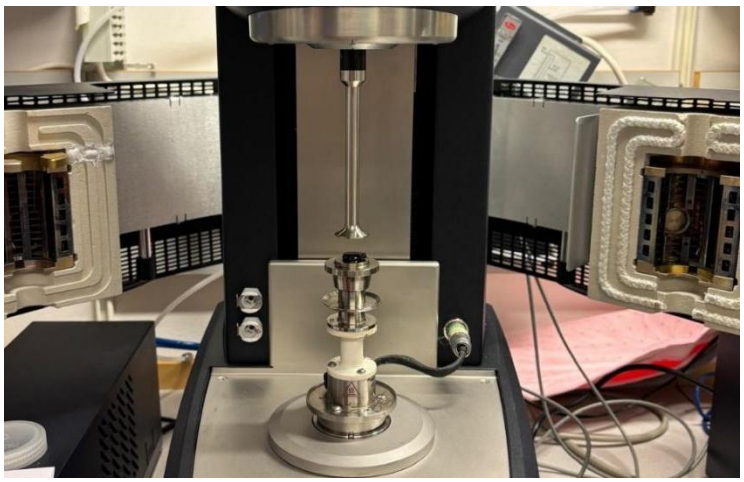


Figure 3: DSR Testing Preparation in KTH Laboratory Instrument

3.3.1 Temperature Sweep

Temperature sweep testing using a Dynamic Shear Rheometer (DSR) is fundamental for analyzing the complex modulus $G*$ and phase angle δ of asphalt binders under varying temperature conditions. This study examined composite systems containing reclaimed asphalt pavement (RAP) binder, Base Bitumen (70/100) and 5% tall oil additive with RAP to understand their viscoelastic behavior across different service temperatures. The procedure maintained a constant strain of 0.5% and a frequency of 1.59 Hz. The testing protocols are utilized with systematic temperature ramping of 1 °C per minute to ensure uniform temperature

equilibration within the tall oil-modified samples. Two different parallel plate sizes are included for the testing setup based on temperature ranges. 25 mm diameter and 1 mm gap plate is used for higher temperatures between 30°C to 80°C and smaller 8 mm diameter plates and 2 mm gaps are used for lower temperatures from 30°C to -10°C to ensure accurate measurements across varying viscosity conditions.

3.3.2 Frequency Sweep

Frequency sweep testing in the dynamic shear rheometer is used to characterize asphalt binder behavior across a wide range of loading rates. It is simulated traffic from slow to high speeds with the frequencies chosen to represent different traffic conditions. As per AASHTO Provisional Standard, "The specified DSR oscillation rate of 10 radians/second (1.59 Hz) is meant to simulate the shearing action corresponding to a traffic speed of about 55 mph (90 km/hr)". In this case, 22 change of frequency is considered (Table 4) to simulate different speeds for several traffic standards according to the Swedish context.

Frequency sweep tests are performed at a fixed temperature range (-10°C to 80°C) as mentioned on section 3.3.1. The oscillation frequency is varied from about 0.1 rad/s up to 20 rad/s representing highway traffic speeds. It is considered to evaluate how complex modulus $G*$ and phase angle δ change with loading rate. The test is conducted maintaining 0.5% shear strain within the linear viscoelastic region (LVE) which is established by prior amplitude sweeps. So, the measured response represented true material properties.

Table 4: Considered Frequency Interval for Testing

SL	Frequency (Hz)	SL	Frequency (Hz)	SL	Frequency (Hz)
1	0.1	9	0.753	16	4.4
2	0.129	10	0.969	17	5.66
3	0.166	11	1.25	18	7.29
4	0.213	12	1.6	19	9.38
5	0.274	13	2.06	20	12.1
6	0.353	14	2.66	21	15.5
7	0.454	15	3.42	22	20
8	0.585				

3.4 Master Curve Construction

For the investigation of the rheological properties of asphalt binders, the construction of a master curve is commonly used to consider a wide range of temperatures and frequencies. In the construction process, the Time–Temperature Superposition Principle (TTSP) is a useful method in asphalt binder characterization that consolidates viscoelastic data from various temperatures into a single predictive master curve (Guo et al., 2018). TTSP states that temperature only shifts the material's response left or right horizontally on a logarithmic frequency axis without changing the shape of the curve, so data from different temperatures can be overlapped. This allows you to combine data collected at various temperatures into a single master curve with a unified view (Chen et al., 2017). In this study, the TTSP method is employed for master curve creation. The procedure are started with frequency sweep testing to acquire the complex modulus G∗ and phase angle δ. Then considered reference temperature 20°C was selected to calculate shift factors α_T that align the data. The Williams–Landel–Ferry (WLF) formula is employed to calculate these shift factors are frequently which mentioned on equation 1.

$$log_{10}\alpha_T = -\frac{C_1(T - T_r)}{C_2 + (T - T_r)} \quad \textbf{Eqn. 1}$$

Where T and T_r represents the test and reference temperature during test, while C_1 and C_2 are material constants from non-linear regression using shift factor values from DSR testing.

The Reduced Frequency ($log\ log\ f_r$) is then calculated by adding the shift factor to the measured frequency (Eqn. 2).

$$log(f_r) = log\ log\ (f)\ + log\ log\ (\alpha_T) \quad \textbf{Eqn. 2}$$

After receiving reduced frequency data as a part of master curve input data on horizontal axis, the Sigmoidal function (Eqn. 3) is used for receiving complex modulus G* value. G* value is used on vertical axis and with this data, the master curve is constructed.

$$log\ log\ (G^*) = \delta + \frac{\alpha}{1 + e^{\beta + \gamma . log(f_r)}} \quad \textbf{Eqn. 3}$$

G* is the complex modulus in Pa, $log(f_r)$ is the reduced frequency in rad/s, and α, β, γ, and δ are coefficients computed using non-linear regression using DSR testing values. These parameters assist in defining the limits, form, and position of the final master curve.

3.5 Cole-Cole Diagram

The Cole–Cole representation methodology involves plotting the loss modulus G″ against the storage modulus G′ from viscoelastic measurements conducted under controlled oscillatory conditions which is completed by DSR testing. This testing method removes specific temperature or frequency scales from the axes which making it much easier to visualize exactly how energy is distributed between elastic storage and viscous dissipation. The storage and loss modulus are derived from the complex modulus G* using the following relationships mentioned in equation 4 and 5 accordingly.

$$G' = G^* + cos\delta \quad \textbf{Eqn. 4}$$

$$G'' = G^* + sin\delta \quad \textbf{Eqn. 5}$$

Where δ represents the phase angle between applied stress and resulting strain. These equations directly connect to mechanical behavior—the cosine term captures stored (recoverable) energy, while the sine term represents dissipated (non-recoverable) energy per cycle.

3.6 Environmental and Economic Analysis

The principal aspects of this study are to demonstrate that bio-composite asphalt (focused on RAP+5% tall oil) can be an effective replacement for conventional asphalt. To convey this, a concise environmental and economic analysis is performed in cooperation with peer-reviewed literature and technical report data. Environment and economic analysis are provided addition benefits of the reclaimed asphalt with tall oil along with rheological properties. In this case, a comprehensive life cycle assessment (LCA) is not analyzed since specific emission data and indicators are unavailable.

Furthermore, the production processes, sensitivity and expansion of procedure system and their environmental impact have not been discussed due to lack of precision and limited data sources.

The simplified approach is to focus on rheological findings and consider advantages of reusing recycled materials and achieving better mechanical properties enhancement that can contribute to pavement engineering towards sustainability.

Chapter 4

4 Results and Discussion

4.1 Rheological outcome

As discussed in Chapter 3 Section 3.4, to receive the binders rheological behavior, a master curve has been constructed after the laboratory testing. All data received from DSR tester software that convey the raw data from data logger while test was continued. Indeed, the master curve and Cole-Cole curve are demonstrated the characteristics of all binders performance. In this study, these two curve outcomes are considered for resulting performance. Based on the curve results, It has been discussed how RAP with tall oil rejuvenator could be more sustainable to implement on future road construction. For construction of master curve, WLF formula and Sigmoidal model both have been implemented and Table 5 presents all parameter data associated with individual binders.

For the Cole-Cole graph, the Loss modulus and storage modulus data received from DSR tester software raw data sheet which apparently calculated using the formula mention on section 3.5.

Table 5: Parameter values used for all binders in WLF formula and Sigmoidal model

Binder	WLF formula	Sigmoidal model
Base Bitumen (70/100)	$log_{10}\alpha_T = -\frac{13.365(T-T_r)}{107.468+(T-T_r)}$	$log\ log\ (G^*) = (-2.325) + \frac{11.141}{1+e^{(-1.263)+0.35(loglog\ (f_r))}}$
RAP	$log_{10}\alpha_T = -\frac{17.681(T-T_r)}{137.355+(T-T_r)}$	$log\ log\ (G^*) = (-4.075) + \frac{12.976}{1+e^{(-1.682)+0.28(loglog\ (f_r))}}$
RAP with 5% Tall Oil	$log_{10}\alpha_T = -\frac{16.433(T-T_r)}{201.958+(T-T_r)}$	$log\ log\ (G^*) = (-68.13) + \frac{79.318}{1+e^{(-2.575)+0.13(loglog\ (f_r))}}$

4.2 Master Curve

Here, the rheological behavior of all three binders is explained with a master curve demonstration. The resulting complex modulus against reduce frequencies is shown on Figure 4 where the base bitumen (70/100) demonstrates the highest complex modulus G∗ values which reaching approximately 8500 Pa. BB has highest stiffness and rheological profile; best for deformation resistance but less forgiving in terms of flexibility. It's apparently indicating superior mechanical performance compared to both RAP-only and RAP with 5% tall oil modifications. The DSR testing data considered as a base of rheological properties, RAP shows Stiffer and more brittle with higher cracking risk. Adding tall oil to the RAP caused a downward vertical shift (lower G∗) in the master curve across all frequencies. This indicates a significant softening effect of tall oil on the RAP binder. Additionally, the transition zone where the material shifts from elastic to viscous behavior, RAP with tall oil demonstrated great results. The transition zone behavior showed:

- Enhanced flexibility in the RAP + 5% tall oil sample.
- Broader transition zones with smoother slopes rather than abrupt changes.
- Better adaptation from elastic to viscous behavior transitions.

Several rheological indicators can be defined after observation of master curve that point out the advantage of RAP with tall oil. Here, several key indicators is described for binders.

Viscoelastic spectrum: The modification with tall oil enhanced the binder's viscous capacity. It successfully balanced the reduction in stiffness with gained flexibility. The resulting viscoelastic spectrum is better balanced. So, the binder could adjust more effectively to temperature and loading variations and improve overall durability.

Stiffness and Durability: The following table (Table 6) summarizes the outcomes associated with the different stiffness levels observed in the data:

Table 6: Observed stiffness of all binders

Stiffness Level	Observed In	Outcome & Implications
Lowest Stiffness	RAP-only Binder	Correlates with increased susceptibility to thermal cracking under low-temperature conditions, linked to a predominance of brittle behavior.
Highest Stiffness	Base Bitumen	Results in improved resistance to permanent deformation under high-load conditions, ensuring better structural integrity during service.
Balanced Stiffness	RAP + 5% Tall Oil	Achieves a balance between elasticity and viscosity, providing higher resistance to rutting.

Stress Distribution & Absorption: The RAP + 5% Tall Oil binder is the most capable of distributing and absorbing stress throughout its matrix. Its improved viscoelastic properties contribute significantly to overall pavement longevity.

Elastic & Viscous Behaviors: The tall oil-modified RAP demonstrated enhanced elastic recovery during loading cycles while maintaining adequate viscous behavior. It effectively dissipated energy under thermal and mechanical stresses. Overall, tall oil created a more favorable elastic–viscous balance compared to RAP alone which close to similar to the BB.

Low Frequency Analysis: Data at low frequencies represents conditions involving rapid loading contexts such as slow traffic loads or significant thermal gradients. The curve for the RAP + 5% Tall Oil at low frequencies have illustrated a notable reduction in complex modulus values compared to the base bitumen. This highlights its enhanced flexibility which is critical for durability under slow loading conditions.

Under these conditions, the tall-oil-modified binder's softer and more viscous nature becomes important under these conditions:

- Accommodate large deformations.

- Avoid sudden cracking despite reduced stiffness.

This is beneficial under very slow loading and high temperatures where a too stiff binder would rut. While a more flexible binder can deform and recover without permanent damage.

Rutting Resistance: The RAP + 5% tall oil binder achieves optimal balance between elasticity and viscosity, providing higher rutting resistance and demonstrating practical advantages in pavement applications.

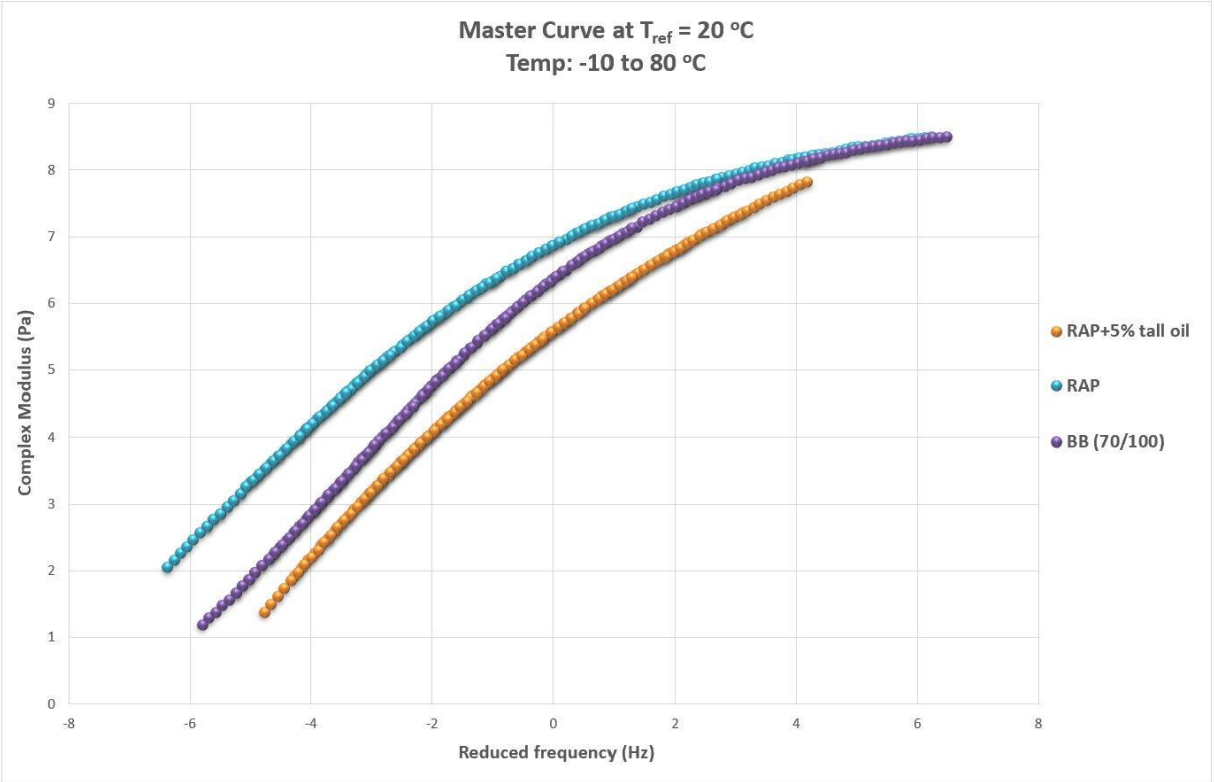


Figure 4: Master Curve for individual binders

The master curve for the RAP + 5% Tall Oil aligns more closely with the Base Bitumen than the untreated RAP does. This reflects the modifier's effectiveness in bridging the performance gap caused by aging, restoring properties closer to the original virgin binder. These findings highlight the significant advantages of incorporating tall oil in RAP binders. Particularly for enhancing performance while maintaining sustainability objectives in pavement engineering applications.

4.3 Cole-Cole Diagram

The Cole-Cole diagram effectively evaluates the homogeneity and compatibility of the asphalt binders through their curve characteristics in this study. The diagram (Figure 5) delineates three specific rheological zones: Viscous Zone, Viscoelastic Transition Zone and Elastic Zone, representing different states of energy behavior. Viscous zone represents energy-dissipative characteristics. Viscoelastic transition zone indicates the material's changing response as it shifts from energy-dissipative (viscous) to energy-storing (elastic) characteristics. The elastic zone represents the state where energy storage is dominant.

The shape of the plots reveals how well the different phases within the binder interact. Base bitumen (70/100) displays a distinct semi-circular arc which indicating excellent compatibility and homogeneous behavior. RAP binder shows significant deviation from the ideal semi-circular pattern that revealing enhanced phase incompatibility due to aging effects. Compared to Rap only, Rap with tall oil demonstrates improved compatibility compared and approaching the BB curve pattern.

As the temperature increases, the storage modulus generally increases, while the loss modulus fluctuates for all curves. These changes indicate shifts in the material's ability to resist deformation and dissipate energy as environmental conditions change. The RAP binder exhibits a stiffer response at lower temperatures, which suggests brittleness that is a common characteristic of aged asphalt. This continuous evolution shows a typical viscoelastic response rather than purely elastic or purely viscous behavior. All binders exhibit viscoelastic relaxation as they have both store and dissipate energy capacities. Considering the phase angle data which highlights shifts in viscoelastic behavior across the temperature spectrum. BB maintained a lower phase angle in the intermediate–high modulus region, consistent with a relatively elastic yet still well-balanced behavior. RAP Binder exhibited a higher phase angle. So, the results reflect increased viscosity and behavior that is described as less elastic in this specific context comparison.

Although all curves seems similar on the Figure 5, one the other way the curves for each binder illustrate distinct viscoelastic behaviors more visual on Figure 6. Each binder curve patterns summarized on Table 7.

Table 7: Binder Cole-Cole Diagram Curve Characteristics

Binder Type	Curve Characteristics	Interpretation
Base Bitumen (BB)	Notably more circular	Suggests superior compatibility.
RAP Only	Straighter and less circular	Indicates a more elastic response attributed to aging.
RAP + 5% Tall Oil	Shifts closer to the BB shape	Shows improved compatibility compared to pure RAP.

RAP with tall oil curves is most similar and closest to the BB curve. The rejuvenator softens the RAP binder and enhances its viscous properties and leading to improved phase compatibility. This change causes the Cole-Cole trajectory of the RAP to resemble that of the original base bitumen, indicating a successful partial restoration of its original rheological properties.

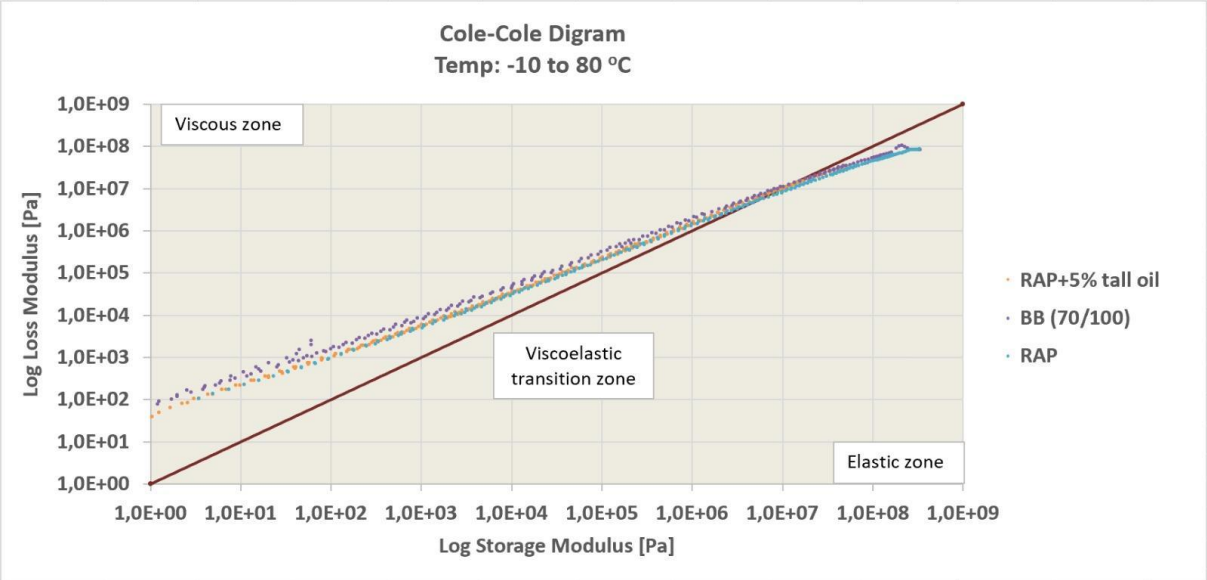


Figure 5: Cole-Cole Diagram of different binders (Log Scale)

The outcomes derived from these Cole-Cole diagrams are highly valuable for material design engineering. The diagram offers critical insights into the compatibility and performance characteristics of various asphalt mixes. Understanding these properties enhances the ability of

engineers to design more robust asphalt solutions capable of withstanding varying environmental conditions and specific application needs.

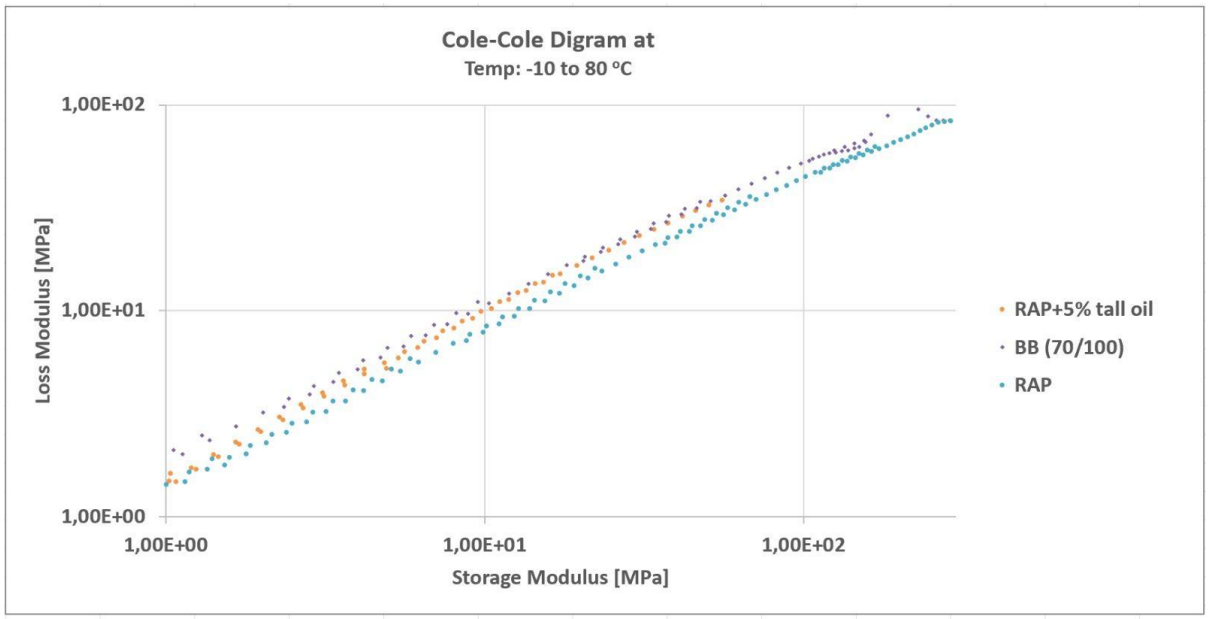


Figure 6: Cole-Cole Diagram of different binders (Non-Log Scale)

In summary, RAP alone is too stiff and elastic-dominant, but adding 5% tall oil improves compatibility and balance, making it a better choice for performance-oriented mixtures.

4.4 Environmental and Economic Advantage

The incorporation of 5% tall oil into recycled asphalt pavement (RAP) binders represents a significant advancement in sustainable pavement engineering, offering compelling economic and environmental advantages. This bio-based approach replaces portions of virgin petroleum bitumen which directly reducing crude oil extraction demands and associated energy consumption (Tarsi, Tataranni and Sangiorgi, 2020).

From an environmental perspective, the tall oil modification delivers substantial greenhouse gas emission reductions of up to 29%, supporting climate change mitigation goals while maintaining pavement performance standards (Sani, Sutanto and Setiadji, 2025). The renewable origin of tall oil contributes biogenic carbon content that effectively displaces fossil-based carbon throughout the pavement's operational lifespan (Cavalli, Wu and Poulikakos, 2024).

The economic benefits are particularly noteworthy, with cost-benefit analysis demonstrating savings of approximately $33.10 USD per tonne compared to conventional neat asphalt binders.

These savings are maximized when tall oil is sourced locally near asphalt production facilities (Lv et al., 2021). The financial advantages extend beyond immediate material costs, encompassing comprehensive life cycle cost reductions of more than 25% in appropriate applications (Sani, Sutanto and Setiadji, 2025).

The methodology significantly improves RAP recyclability by allowing rejuvenated mixtures to be processed with substantially reduced virgin material requirements. This creates a sustainable cycle of cost reductions and resource conservation. For instance, significant reducing use of landfills, leads to long-term economic benefits that extend well beyond initial construction phases (Tarsi, Tataranni and Sangiorgi, 2020; Sani et al., 2025). The approach not only supports emission reduction targets but also extends pavement service life, resulting in lower long-term maintenance expenditures and improved infrastructure sustainability (Cavalli, Wu and Poulikakos, 2024).

Chapter 5

5 Conclusion and Recommendations

This thesis fundamentally focus on binder level performance incorporating with rejuvenator. The analysis explores the influence of bio additives on the binder stiffness, viscoelastic and temperature sensitivity of several weather condition. The demonstration of sustainable alternatives of conventional binders examined through rheological analysis along with considering the environment and economy.

5.1 Conclusion

The incorporation of 5% tall oil as a rejuvenator into reclaimed asphalt pavement (RAP) demonstrates significant improvements in mechanical properties when compared to both untreated RAP and virgin bitumen (70/100). Rheological analysis has revealed that tall oil-modified RAP binders exhibited lower complex modulus across all frequencies in their unaged state, resulting in a softer and more flexible binder compared to virgin bitumen. Moreover, this bio-based additive replenishes maltene fractions, improving the viscoelastic balance of the RAP binder while enhancing flexibility and maintaining rutting resistance. The rejuvenation effect proves particularly beneficial after aging processes as RAP with 5% tall oil maintained superior performance by effectively restoring stiffness levels closer to those of aged virgin binder. At the same time untreated RAP experienced greater modulus increases that indicated heightened brittleness. Beyond mechanical enhancements, the tall oil addition positively influenced chemical properties, enhanced adhesion and reduced moisture susceptibility factors critical for long-term pavement durability. The molecular structure of the RAP binder are favorably altered through tall oil incorporation, for instance leading to improved elastic properties and a more stable microstructure.

From an environmental standpoint, tall oil promotes circular economy practices by utilizing industrial byproducts and contributes to lower greenhouse gas emissions and energy consumption. Economically, it can reduce the demand for virgin binders and extend pavement service life. Thereby, it can also decrease maintenance costs while its implementation at an industrial scale is feasible with appropriate temperature control and quality assurance measures.

Overall, tall oil-modified RAP binders emerge as a sustainable and practical alternative in pavement engineering, combining improved durability, environmental responsibility and economic efficiency.

5.2 Recommendations for Future Research

Based on the identified gaps and limitations, the following research priorities are recommended:

Optimization of Tall Oil Percentage

Future research should concentrate on optimizing tall oil dosage in RAP binders to balance improved low-temperature flexibility and fatigue life. Although a 5 % addition has shown promising rheological benefits, adjusting this concentration up or down may significantly alter performance under different climatic and traffic conditions because the softening effect of tall oil does not increase linearly with dosage. Future work should therefore implement comprehensive rheological testing for instance DSR testing across multiple temperatures and frequencies, MSCR and LAS testing for rutting and fatigue characterization across a systematic range of concentrations. Those tests require to identify blends that provide the best overall performance. In this case, multiple RAP contents (e.g., 10%, 25%, 40% and 50%) can be tested to establish trends. At the same time, integrating quality control, cost considerations, and sustainability criteria into the selection process should be considered. Field trials and long-term environmental performance assessments are also needed to validate laboratory findings and to develop practical guidelines for the effective and sustainable use of tall oil in pavement applications.

Integration with Other Bio-based Additives

Integrating oil with other bio-based additives would be enhance the performance and durability of RAP binders. Several studies are ongoing to better understand the performance of bio additive binders. To emphasis, tall oil could be one of the effective additives which can provide revolutionary alternatives. Now requires more comprehensive study, for example Synergistic formulations that combine tall oil with lignin could exploit lignin's structural reinforcement to counteract tall oil's softening effect at high temperatures. This combination is improving rutting resistance while maintaining rejuvenation benefits. Several Studies should also examine the use of epoxidized vegetable oils whose covalent bonding with binder components can complement

tall oil's maltene restoration and potentially increase elasticity and network stability. Further investigations into animal waste-derived bio-binders may reveal additional advantages when used alongside tall oil such as improved aggregate wetting and enhanced low-temperature strain capacity. Another requirement of comprehensive study like tall oil incorporated with small amounts of waste cooking oils and polymeric additives could enable finely tuned formulations that preserve mechanical integrity, optimise sustainability and allow performance to specific climatic and traffic conditions.

Long-term Field Performance Studies

Future research on 5% tall oil-modified RAP binders should focus on testing durability and environmental impact across various climates and traffic levels in field. By constructing test sections in both hot and cold regions, researchers can directly compare these modified binders against standard options. Regular mechanical testing and visual inspections are necessary to monitor performance issues like cracking and determine the pavement's overall lifespan.

Research should also look at how tall oil improves moisture resistance and flexibility at low temperatures and whether these improvements actually result in fewer moisture problems and thermal cracks over time.

Life Cycle Assessment

This study lacks of comprehensive LCA of tall oil-modified RAP systems including cradle-to-gate emissions, transportation impacts, mixing energy and end-of-life considerations. LCA needs to implement to understand precise environmental effects and benefits which suggests a large area of research requirement.

Chemical Characterisation

Detailed chemical analysis of tall oil (for example, FTIR, GC-MS, elemental analysis, and acid value) is required to enable structure-property relationships to be established and compared with other bio-oils.

Bibliography

- Alattieh, S.A., Al-Khateeb, G.G. and Zeiada, W. (2021) 'Evaluation of low- and intermediate-temperature performance of bio oil-modified asphalt binders', *Sustainability*, 13(7), pp. 4039. Available at: https://doi.org/10.3390/su13074039.
- Aro, T. and Fatehi, P. (2017) 'Tall oil production from black liquor: Challenges and opportunities', *Separation and Purification Technology*, 175, pp. 469–480. Available at: https://doi.org/10.1016/j.seppur.2016.10.027.
- Barco Carrión, A.J. del, *et al*. (2023) 'Rheological investigation on the ageing performance of bio-recycled asphalt binders and mixtures', *Road Materials and Pavement Design*, 24(sup1), pp. 538–553. Available at: https://doi.org/10.1080/14680629.2023.2181010.
- Cavalli, M.C., Zaumanis, M., Mazza, E., Partl, M.N. and Poulikakos, L.D. (2018) 'Effect of ageing on the mechanical and chemical properties of binder from RAP treated with bio-based rejuvenators', *Composites Part B: Engineering*, 141, pp. 174–181. Available at: https://doi.org/10.1016/j.compositesb.2017.12.060.
- Cavalli, M.C. (2018) *Exploring fundamental aspects of bio-modified reclaimed asphalt binder*. DISS. ETH NO. 25230. Doctoral thesis, ETH Zurich. Available at: https://doi.org/10.3929/ethz-b-000271594.
- Cavalli, M.C., M.C. *et al.* (2021) 'Surface nanomechanical properties of bio-modified reclaimed asphalt binder', *Road Materials and Pavement Design*, 22(6), pp. 1407–1423. Available at: https://doi.org/10.1080/14680629.2019.1691042.
- Cavalli, M.C., Wu, W. and Poulikakos, L. (2024) 'Bio-based rejuvenators in asphalt pavements: A comprehensive review and analytical study', *Journal of Road Engineering*, pp. 282–291. Available at: https://doi.org/10.1016/j.jreng.2024.04.007.
- Chen, C. *et al.* (2017) 'Preliminary investigation of bioadvantaged polymers as sustainable alternatives to petroleum-derived polymers for asphalt modification', *Materials and* Structures, 50:225. Available at: https://doi.org/10.1617/s11527-017-1097-4.
- Chen C., Lu J., Ma T., Zhang Y., Gu L. and Chen X. 2023 'Applications of vegetable oils and their derivatives as bio-additives for use in asphalt binders: A review', *Construction*

and Building Materials, 383, p. 131312. Available at: https://doi.org/10.1016/j.conbuildmat.2023.131312.

- Dony, A., Colin, J., Bruneau, D., Drouadaine, I. and Navaro, J. (2013) 'Reclaimed asphalt concretes with high recycling rates: changes in reclaimed binder properties according to rejuvenating agent', *Construction and Building Materials*, 41, pp. 175–181. Available at: https://doi.org/10.1016/j.conbuildmat.2012.11.031.
- Elnaml, I., Mohammad, L.N., Baumgardner, G.L., *et al.* (2024) 'Influence of petroleum-based and bio-derived recycling agents on high-RAP asphalt mixtures performance', *Buildings*, 14, p. 567. Available at: https://doi.org/10.3390/buildings14030567.
- Elnaml, I., Mohammad, L.N., Baumgardner, G., *et al.* (2024) 'Sustainability of asphalt mixtures containing 50% RAP and recycling agents', *Recycling*, p. 85. Available at: https://doi.org/10.3390/recycling9050085.
- Eurobitume (2025) 'The eurobitume life cycle assessment 4.0 for bitumen'. Available at: https://www.eurobitume.eu.
- Fini, E.H. *et al.* (2012) 'Partial replacement of asphalt binder with bio-binder: Characterisation and modification', *International Journal of Pavement Engineering*, 13(6), pp. 515–522. Available at: https://doi.org/10.1080/10298436.2011.596937.
- Fini, E.H. and Hajikarimi, P. (2025a) 'Sources and production', *in Bio-based and bio-inspired pavement construction materials*. Available at: https://doi.org/10.1016/B978-0-443-18634-9.00006-4.
- Fini, E.H. and Hajikarimi, P. (2025b) 'Environmental assessment and pavement lifecycle', *in Bio-based and bio-inspired pavement construction materials*. Available at: https://doi.org/10.1016/B978-0-443-18634-9.00003-9.
- Fini, E.H. and Hajikarimi, P. (2025c) 'Numerical modeling and simulation', *in Bio-based and bio-inspired pavement construction materials*. Available at: https://doi.org/10.1016/B978-0-443-18634-9.00011-8.
- Fini, E.H. and Hajikarimi, P. (2025d) 'Supply chain management and the circular economy', *in Bio-based and bio-inspired pavement construction materials*. Available at: https://doi.org/10.1016/B978-0-443-18634-9.00004-0.
- Fuhaid, A.A., Lu, Q. and Luo, S. (2018) 'Laboratory evaluation of biobased epoxy asphalt binder for asphalt pavement', *Journal of Materials in Civil Engineering*. Available at: https://doi.org/10.1061/(ASCE)MT.1943-5533.0002383.

- Gao, J., Guo, G., Wang, H., Jin, D., Bi, Y. and Jelagin, D. (2025) 'Research progress of bio-asphalt towards green pavement development: Preparation, properties, and mechanism', *Fuel*, 381, p. B. Available at: https://doi-org.focus.lib.kth.se/10.1016/j.fuel.2024.133409.
- Gaudenzi, E. *et al.* (2021) 'Performance assessment of asphalt mixture produced with a bio-based binder', *Materials*, 14(4). Available at: https://doi.org/10.3390/ma14040918.
- Gowda, S. et al. (2024) 'Analysis of rutting behaviour of recycled asphalt binder and rejuvenated recycled asphalt binder by multiple stress creep recovery (MSCR) test', *IOP Conference Series: Earth and Environmental Science*, 1326, p. 012065. Available at: doi:10.1088/1755-1315/1326/1/012065.
- Guo, G. *et al.* (2024) 'Study on the storage stability and rheological property of bio-oil/lignin composite-modified asphalt', *Polymers*, 16, p. 2484. Available at: https://doi.org/10.3390/polym16172484.
- Guo, W. *et al.* (2018) 'Evaluating the effect of hydrophobic nanosilica on the viscoelasticity property of asphalt and asphalt mixture', *Materials*. Available at: https://doi.org/10.3390/ma11112328.
- Imaninasab, R., Loria-Salazar, L. and Carter, A. (2024) 'Rheological analysis of blended vs. Recovered asphalt binders in rejuvenated mixtures with high reclaimed asphalt pavement', *Mechanics of Time-Dependent Materials*, pp. 3049–3068. Available at: https://doi.org/10.1007/s11043-024-09711-6.
- Jawad, M., Fini, E.H. and Abu-Lebdeh, T.M. (2017) 'Investigating the efficacy of a bio-based modifier to improve the rheological properties of recycled asphalt shingles', *American Journal of Engineering and Applied Sciences*, 10 (1), pp. 165–174. Available at: 10.3844/ajeassp.2017.165.174.
- Jiao, X. *et al.* (2024) 'Study on the compatibility of SBR and asphalt base based on molecular simulation', *Materials*, 17, p. 1175. Available at: https://doi.org/10.3390/ma17051175.
- Kowalski, K.J. *et al.* (2017) 'Thermal and fatigue evaluation of asphalt mixtures containing RAP treated with a bio-agent', *Applied Sciences*, p. 216. Available at: https://doi.org/10.3390/app7030216.
- Kumar, V. and Aggarwal, P. (2023) 'Characteristics of waste oil- rejuvenated RAP bitumen: An experimental study', *Jordan Journal of Civil Engineering*, 17(3). Available at: https://doi.org/10.14525/JJCE.v17i3.07.

- Liu, G. *et al.* (2022) 'Study on the conventional performance and microscopic properties of PPA/SBS-modi02ed bio-mixed asphalt', *Materials*, p. 4101. Available at: https://doi.org/10.3390/ma15124101.
- Lv, S. *et al.* (2021) 'Preparation and performance of polyphosphoric acid/bio-oil composite–modified asphalt containing a high content bio-oil', *Journal of Materials in Civil Engineering*. Available at: https://doi.org/10.1061/(ASCE)MT.1943-5533.0004124.
- Moins, B., Hernando, D., Buyle, M., France, C., Bergh, W.V.D. and Audenaert, A. (2022) 'On the road again! An economic and environmental break-even and hotspot analysis of reclaimed asphalt pavement and rejuvenators, *Resources, Conservation & Recycling*, 177, p. 106014. Available at: https://doi.org/10.1016/j.resconrec.2021.106014.
- Nazzal, M.D. *et al.* (2014) 'Multiscale evaluation of the composite asphalt binder in high –reclaimed asphalt pavement mixtures', *Journal of Materials in Civil Engineering*. Available at: https://doi.org/10.1061/(ASCE)MT.1943-5533.0000825.
- Nguyen, L.N. *et al.* (2024) 'Effects of rejuvenators and aging conditions on the properties of blended bitumen and the cracking behavior of hot asphalt mixtures with a high RAP content', *Proceedings of the Institution of Mechanical Engineers, Part L: J Materials: Design and Applications*. Available at: https://doi.org/10.1177/14644207241236901.
- Nguyen, L.N. *et al.* (2025) 'Effects of lab-prepared soybean oil-based rejuvenator on high reclaimed asphalt pavement content asphalt mixtures', *Proceedings of the Institution of Mechanical Engineers, Part L: J Materials: Design and Applications*. Available at: https://doi.org/10.1177/14644207251384742.
- Ongel, A. and Hugener, M. (2015) 'Impact of rejuvenators on aging properties of bitumen', *Construction and Building Materials*, pp. 467–474. Available at: https://doi.org/10.1016/j.conbuildmat.2015.07.030.
- Pei, X. and Fan, W. (2021) 'Effects of amorphous poly alpha olefin (APAO) and polyphosphoric acid (PPA) on the rheological properties, compatibility and stability of asphalt binder', *Materials*, 14. Available at: https://doi.org/10.3390/ma14092458.
- Riaz, A. and Hafeez, I. (2022) 'Characterizing the Bituminous Material by Using Graphical Techniques', *International Journal Research in Applied Science & Engineering (IJRASET)*, 10(4). Available at: https://doi.org/10.22214/ijraset.2022.41780.
- Samara, M. *et al.* (2022) 'Performance evaluation and characterization of extracted recycled asphalt binder with rejuvenators', *Transportation Research Record*, 2676(12), pp. 714–727. Available at: https://doi.org/10.1177/03611981221097091.

- Sani, N.M., Sutanto, H. and Setiadji, B.H. (2025) 'A bibliometric analysis of reclaimed asphalt pavement for green infrastructure', *E3S Web of Conferences*, p. 11002. Available at: https://doi.org/10.1051/e3sconf/202565211002.
- Tarsi, G., Tataranni, P. and Sangiorgi, C. (2020) 'The challenges of using reclaimed asphalt pavement for new asphalt mixtures: A review', *Materials*. Available at: doi:10.3390/ma13184052.
- Toth, C., Petho, L. and Rosta, S. (2023) 'Rheological characterisation of bituminous binder blends for the design of asphalt mixes with high recycled asphalt content', *ACTA TECHNICA JAURINENSIS*, 16(2), pp. 62–74. Available at: https://doi.org/10.14513/actatechjaur.00694.
- Wang, D. *et al.* (2019) 'An alternative experimental method for measuring the low temperature rheological properties of asphalt binder by using 4 mm parallel plates on dynamic shear rheometer', *Transportation Research Record*, 2673(3), pp. 427–438. Available at: https://doi.org/10.1177/0361198119834912.
- Wang, W. *et al.* (2020) 'Research on rheological properties of high-percentage artificial RAP binder with WMA additives', *Advances in Materials Science and Engineering*, Vol. 2020, Article ID 1238378, p. 24. Available at: https://doi.org/10.1155/2020/1238378.
- Wu, Q. and Hou, G. (2024) 'The impact of bio-oil on the structure, rheology, and adhesion properties of lignin-modified asphalt', *Materials Research Express*, 11, p. 065101. Available at: https://doi.org/10.1088/2053-1591/ad55ad.
- Xu, H. *et al.* (2023) 'Investigation into rheological behavior of warm-mix recycled asphalt binders with high percentages of RAP binder', *Materials*, 16, p. 1599. Available at: https://doi.org/10.3390/ma16041599.
- Yaseen, G. *et al.* (2022) 'Performance evaluation of aged asphalt pavement binder through rejuvenators', *Sustainability*, 14, p. 14557. Available at: https://doi.org/10.3390/su142114557.
- Yu X., Zaumanis M., Santos S.D., Poulikakos L. D. 2014. Rheological, microscopic, and chemical characterization of the rejuvenating effect on asphalt binders. *The Science and Technology of Fuel and Energy* 135:162-171. Available at: https://doi-org.focus.lib.kth.se/10.1016/j.fuel.2014.06.038
- Yusoff, N.I.M., Hainin, M.R., Mounier, D. and Airey, G.D. (2013) 'Use of inter-conversion equations on bituminous binder data', *Sains Malaysiana*, 42(11) pp. 1647–1654. Available at:

https://www.ukm.my/jsm/english_journals/vol42num11_2013/vol42num11_2013pg1647-1654.html.

- Zhang, J. et al. (2020) 'Effect of different viscous rejuvenators on chemical and mechanical behavior of aged and recovered bitumen from RAP', Construction and Building Materials, 239, p. 117755. Available at: https://doi.org/10.1016/j.conbuildmat.2019.117755.
- Zhu, J. et al. (2023) 'Durability assessment of bio-extended bituminous binders by rheological characterisation after long-term ageing', *Materials*, 16(4), p. 1599. Available at: https://urn.kb.se/resolve?urn=urn:nbn:se:vti:diva-19888.
- Zhu, J. *et al.* (2025) 'Long-term performance of asphalt mixtures containing bio-extended bituminous binders: Laboratory investigation and accelerated pavement testing', *Road Materials and Pavement Design*, pp. 135–152. Available at: https://doi.org/10.1080/14680629.2025.2482845.